

Class-II joint source-potential reduction and local Bellman no-go

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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Purpose and scope

The broad joint gate is reduced but not closed. The exact conditional source-potential formula is now fixed. On the A13 opposite-corner carrier, the coefficient-variation source has the same leading amplitude and sign as the A11 frozen derivative source. The exact fast-phase principal symbol therefore has twice the frozen source, and its square has a factor four; the registered modulated carrier inherits this asymptotically. A coefficient-one potential-increment estimate and, more generally, a finite bank of local Class-II/quartic/sextic polynomial terms with bounded coefficients and cutoff-summable positive replenishments and scalar transfer errors fail on this carrier.

This does *not* refute the full action. Once a Cameron–Martin cost and a retained past sextic are present, the registered carrier is strongly subcritical. A universal one-shell frozen-source inequality and a mixed Hardy/Riesz lemma make that feasibility quantitative. What remains is a single cutoff-uniform adapted-control theorem which spends the Cameron–Martin energy only once across all scales. The package remains T4; it does not prove the A7 Nelson bound, an interacting measure, T5, T6, or T7.

The authority is the hash-pinned A1 production functional on the $L = 16$ torus, three complex fields in the six-real convention, fixed $\rho_{\text{floor}} = 10^{-12}$, $Y = 1$, the common real-even sharp regulator, and the A10–A13 strict dyadic rectangular-cube filtration. Write

$$U(z) = \int_{\mathbb{T}_L^3} \left(\frac{\lambda}{4} |z|^4 + \frac{\gamma}{6} |z|^6 \right) dx, \quad \lambda = -0.43, \quad \gamma = 1.62. \quad (1.1)$$

2. Exact conditional joint formula

At shell j , set $x = \phi_{j-1}$, $z = x + K_j \xi_j$, and let ξ_j be the standard real Gaussian innovation. With

$$A_j = M_{B(x)}^{1/2} G_j, \quad T_j = A_j^* A_j, \quad r_j = M_{B(x)}^{1/2} D x, \quad \ell_j = A_j^* r_j, \quad (2.1)$$

the A11 true increment is

$$I_j = \frac{1}{2} \{ \langle \xi_j, T_j \xi_j \rangle - \text{Tr } T_j \} + \langle \ell_j, \xi_j \rangle. \quad (2.2)$$

The coefficient increment is

$$\mathcal{C}_j = q_{B(z)-B(x)}(Dz) - t_{\Gamma_{\leq j}}(B(z) - B(x)), \quad (2.3)$$

and $I_j + \mathcal{C}_j = V_j - V_{j-1}$ exactly.

For $q = p\theta$, put

$$R_q = (I + qT_j)^{-1}, \quad \mu_q = -qR_q \ell_j. \quad (2.4)$$

Let $\kappa = 0$ mean that the full local potential is retained and $\kappa = 1$ mean that only its increment is retained. Completing the Gaussian square, without estimating $\mathcal{C}_j$ or U , gives

$$\begin{aligned} & \log \mathbb{E}_j \exp\{-p[\theta I_j + \mathcal{C}_j + U(z) - \kappa U(x)]\} \\ &= p\kappa U(x) - \frac{1}{2} \log \det {}_2(\mathbf{I} + qT_j) + \frac{q^2}{2} \langle \ell_j, R_q \ell_j \rangle \\ & \quad + \log \mathbb{E}_{\eta \sim N(\mu_q, R_q)} \exp\{-p[\mathcal{C}_j(x, \eta) + U(x + K_j \eta)]\}. \end{aligned} \quad (2.5)$$

Equation (2.5) fixes all signs. Both the regularised-determinant term and the noncentral source term are nonnegative. The sextic supplies the negative large-field exponent. The sign-indefinite $\mathcal{C}_j$ cannot be dropped.

The primary executable checks (2.5) pointwise at random matrix inputs. The non-importing route checks the integrated identity by independent Gauss–Hermite quadrature with a nonzero quartic–sextic test potential.

3. Why the complete local source has a factor four

For the symmetric opposite-corner carrier write, after removal of the fast phases,

$$x = (\psi, \bar{\psi}, 0), \quad D_i x = (iN\psi, -iN\bar{\psi}, 0). \quad (3.1)$$

The exact A13 Fierz identity gives

$$B(x)D_i x = i2\beta_{\text{op}}N(\psi|\psi|^2, -|\psi|^2\bar{\psi}, 0), \quad (3.2)$$

where

$$\beta_{\text{op}} = 4(a + 2b + c) = 0.0423749999999894. \quad (3.3)$$

Consequently the raw Class-II density is

$$q_B(x) = 6\beta_{\text{op}}N^2|\psi|^4. \quad (3.4)$$

Let $y = (\eta, \bar{\eta}, 0)$ be a same-corner shell amplitude variation. Differentiating only the derivative slots of $q_B = \frac{1}{2} \sum_i (D_i x)^T B(x) D_i x$ produces the A11 frozen source. Differentiating the explicit expression (3.4) produces twice that result. Therefore

$$\frac{1}{2} \sum_i (D_i x)^T DB(x)[y] D_i x = \sum_i (D_i x)^T B(x) D_i y, \quad (3.5)$$

and the coefficient-slot source equals the frozen derivative-slot source. Thus

$$\boxed{\ell_{\text{joint}} = 2\ell_{\text{frozen}}, \quad \|\ell_{\text{joint}}\|^2 = 4\|\ell_{\text{frozen}}\|^2.} \quad (3.6)$$

The two executables verify (3.5) directly in this homogeneous fast-phase principal-symbol fixture: one uses the compact Fierz matrix, the other the three Pauli currents. For the actual degree- K modulated carrier, envelope derivatives and the trace/potential terms are not asserted to vanish at finite K ; Section 5 proves that they are lower order in the carrier limit. Thus the factor four is exact for the principal symbol and asymptotic for the registered carrier, not an unqualified finite-shell identity.

4. Terminal and past normalisations are different

For the A13 polynomial f_K and $u_K = P_- f_K$, put

$$B_K = \|f_K\|_6^6, \quad D_K = \|u_K\|_6^6, \quad C_K = \|P_- (|u_K|^2 u_K)\|_2^2. \quad (4.1)$$

At $K = 65536$, two finite routes give

$$\frac{D_K}{B_K} = 9.7254356828\dots, \quad \frac{C_K}{D_K} = 0.7636937448\dots \quad (4.2)$$

The frozen source normalised by the terminal field is

$$C_{\text{rel}} = \beta_{\text{op}}^2 \left[\left(\frac{D_K}{B_K} \right)^3 - \left(\frac{C_K}{D_K} \right)^3 \right] = 0.9160527283\dots \quad (4.3)$$

The same source normalised by the actual conditional past is only

$$c_{\text{past}} = \beta_{\text{op}}^2 \left[1 - \left(\frac{C_K}{D_K} \right)^3 \right] = 0.0009958485\dots, \quad (4.4)$$

because

$$\frac{\|x\|_6^6}{\|\Phi\|_6^6} = \left(\frac{D_K}{B_K} \right)^3 = 919.8715736\dots \quad (4.5)$$

For the complete local source (3.6), the past-normalised exponent reward at $p = 1.1$ is

$$\frac{p^2}{2} (4c_{\text{past}}) = 0.0024099534\dots \quad (4.6)$$

If the past potential itself is retained, its sextic exponent penalty is

$$\frac{p\gamma}{6} = 0.297, \quad (4.7)$$

leaving $0.2945900466\dots$. This comparison is deliberately labelled *past-normalised*. It is not a comparison with the much smaller terminal field Φ and does not itself prove a global theorem.

5. Asymptotic no-go for the local potential-increment route

Consider a high carrier with amplitude A_N and

$$\tau_N = \frac{A_N^2}{N^2} \longrightarrow 0. \quad (5.1)$$

Then $\|T_j\| = O(\tau_N)$, while the trace-gradient and local-potential gradient are lower by $O(\tau_N)$ relative to the source. The potential Hessian is lower by $O(\tau_N^2)$. Equations (3.6) and (4.3) therefore give the terminal-normalised joint constant

$$C_{\text{joint}} = 4C_{\text{rel}} = 3.6642109131\dots \quad (5.2)$$

The entire production sextic can supply at most

$$\frac{\gamma}{3p} \leq \frac{\gamma}{3} = 0.54. \quad (5.3)$$

Hence the coefficient-one conditional estimate

$$\sup_{j,x} \log \mathbb{E}_j \exp\{-p[I_j + \mathcal{C}_j + U(z) - U(x)]\} < \infty \quad (5.4)$$

is false for every $p \geq 1$.

The same scaling rules out the following precisely scoped class. A *finite-bank local Bellman barrier* is allowed to use bounded coefficients of q_B , $\|x\|_4^4$, and $\|x\|_6^6$, but its positive coefficient replenishments and scalar transfer errors must be summable in the cutoff. Along a subsequence their positive increments tend to zero. Choose

$$\tau_j = \max\{\sqrt{\delta_j}, N_j^{-1}\}, \quad (5.5)$$

where δ_j is the available positive coefficient replenishment. Then $T_j = O(\tau_j)$, the bank contribution divided by A_j^6 is $O(\delta_j/\tau_j) = o(1)$, and all local transport corrections are $o(1)$, while (5.2) persists. Therefore no such barrier closes the production budget. Nonlocal value functions, adapted-control costs, smooth covariance flows, and redesigned actions are expressly outside this no-go.

This result is registered as

NG-2026-07-21-A13-LOCAL-BELLMAN-BARRIER.

6. A mixed Hardy/Riesz lemma

The relative-phase carrier nevertheless admits a useful analytic enclosure. Use normalised Haar measure on $\mathbb{T}$. Let $g = P_{<0}f$ and let $Q = P_{\leq 0}$ be used only on products. Since $f - g$ has nonnegative modes and $\bar{g}$ has strictly positive modes,

$$Q(f\bar{g}) = Q(|g|^2). \quad (6.1)$$

For the real function $u = |g|^2$, Fourier symmetry gives

$$\|Qu\|_2^2 = \frac{1}{2}\{\|u\|_2^2 + |\hat{u}(0)|^2\} \geq \frac{1}{2}\|u\|_2^2. \quad (6.2)$$

Therefore

$$\|g\|_4^4 \leq 2\|Q(f\bar{g})\|_2^2 \leq 2\|f\|_6^2\|g\|_3^2. \quad (6.3)$$

The sharp Riesz norm $\|P_{<0}\|_{3 \rightarrow 3} = \csc(\pi/3) = 2/\sqrt{3}$ and $\|f\|_3 \leq \|f\|_2^{1/2}\|f\|_6^{1/2}$ imply

$$\boxed{\frac{\|P_{<0}f\|_4^8}{\|f\|_2^2\|f\|_6^6} \leq \frac{64}{9}}. \quad (6.4)$$

For the three-dimensional tensor frozen carrier this yields the coherent-ray threshold

$$\alpha b \geq \frac{p\beta_{\text{op}}^2}{8Y} \left(\frac{64}{9}\right)^3 = 0.0887839605\dots \quad (6.5)$$

at $p = 1.1$. The illustrative allocation $\alpha = 0.45$, $b = 0.21$ has $\alpha b = 0.0945$, a positive but narrow margin for the *frozen* source. The complete local source has the separate factor four from (3.6), raising this coarse threshold to $0.355135842\dots$; thus (6.4) alone does not close the joint gate. This distinction prevents double counting the two components, three axes, or six-real normalisation already present in the coefficient of (6.5).

The coherent-ray allocation in (6.5) and the one-shell Young allocation in the next section are alternative diagnostics. They are never added together or charged to the same entropy/sextic reserve.

7. Universal one-shell Cameron–Martin/sextic crossover

Let

$$H_A(x) = \langle x, Ax \rangle, \quad L_6(x) = \int |x|^6.$$

A13 gives the exact resolvent rebate

$$\langle \ell, (I + pT)^{-1} \ell \rangle \leq \frac{1}{p} \|r\|_2^2. \quad (7.1)$$

The sharp matrix bound $B(x) \leq \beta_{\text{op}} |x|^2 I$ gives

$$\|r\|_2^2 \leq \beta_{\text{op}} \int |x|^2 |Dx|^2. \quad (7.2)$$

Integration by parts and the production symbol lower bound $A(k) \geq c_{\text{sym}} \langle k \rangle^4 I$ yield

$$\begin{aligned} \int |x|^2 |Dx|^2 &\leq \int |x|^2 x \cdot (-\Delta x) \\ &\leq L_6(x)^{1/2} \|\Delta x\|_2 \leq c_{\text{sym}}^{-1/2} L_6(x)^{1/2} H_A(x)^{1/2}, \end{aligned} \quad (7.3)$$

where $c_{\text{sym}} = 0.10836453058711468$. Young's inequality proves, for every $\vartheta > 0$,

$$\boxed{\frac{p^2}{2} \langle \ell, (I + pT)^{-1} \ell \rangle \leq \vartheta H_A(x) + \frac{p^2 \beta_{\text{op}}^2}{16 c_{\text{sym}} \vartheta} L_6(x).} \quad (7.4)$$

At $p = 1.1$ and $\vartheta = 0.45$, the frozen-source sextic coefficient in (7.4) is $0.0027847431\dots$. On the registered factor-four carrier it is $0.0111389725\dots$. The production potential obeys

$$\frac{\lambda}{4} \rho^2 + \frac{\gamma}{6} \rho^3 \geq \frac{\gamma}{12} \rho^3 - \frac{|\lambda|^3}{3\gamma^2}, \quad (7.5)$$

so the half-sextic exponent coefficient is $p\gamma/12 = 0.1485$ and the registered factor-four margin is $0.1373610275\dots$.

Equation (7.4) is a genuine all-field, all-shell frozen-source theorem. It is not summable as written: x contains Gaussian past noise, and paying $H_A(x)$ at every shell recreates the A11 $O(N)$ base-Gaussian divergence. This is the exact point at which a full-action adapted-control argument is needed.

8. Exact backward recursion and surviving theorem

Let $S_j = V_j + U$ and

$$J_j(x, \xi) = S_j(x + K_j \xi) - S_{j-1}(x) = I_j + \mathcal{C}_j + U(x + K_j \xi) - U(x). \quad (8.1)$$

The exact backward value function is

$$W_J = 0, \quad W_{j-1}(x) = -\frac{1}{p} \log \mathbb{E}_j e^{-p[J_j(x, \xi_j) + W_j(x + K_j \xi_j)]}. \quad (8.2)$$

Equivalently, the finite-dimensional Boue–Dupuis formula gives

$$-\frac{1}{p} \log \mathbb{E} e^{-p S_J} = \inf_v \mathbb{E} \left[S_J(X + h(v)) + \frac{1}{2p} \sum_j \|v_j\|_2^2 \right]. \quad (8.3)$$

Combining (7.5) with (8.3) reduces positive closure to one theorem:

For every progressive control v , uniformly in the cutoff, prove

$$\mathbb{E}V_j^{\text{ren}}(X + h(v)) \geq -\epsilon_6 \mathbb{E}\|X + h(v)\|_6^6 - \epsilon_v \mathbb{E} \sum_j \|v_j\|_2^2 - C, \quad (8.4)$$

with

$$\epsilon_6 < 0.135, \quad \epsilon_v < \frac{1}{2p}. \quad (8.5)$$

At $p = 1.1$, the second threshold is $0.454545\dots$. The proof must retain the A9 Hilbert–Schmidt cancellation for Gaussian fluctuations, use the saturated estimate only on the Cameron–Martin drift, spend that drift energy once across the scale sum, and verify that no new interaction counterterm is generated. Once (8.4) holds, (8.3) and (7.5) give the cutoff-uniform Nelson/log-Laplace bound directly.

9. Devil’s-advocate review

1. **Objection: the factor four was declared rather than derived. Dismissed.** Equation (3.5) derives amplitude doubling from the raw principal-symbol density. Independent compact-Fierz and direct-Pauli finite fixtures verify the coefficient and derivative slots separately, while Section 5 controls the lower-order carrier corrections asymptotically.
2. **Objection: the 0.297 sextic comparison uses the terminal field. Valid with mitigation.** It does not. Section 4 explicitly labels it as a retained *past* potential comparison. Terminal normalisation restores the factor $919.87\dots$ and gives the local no-go.
3. **Objection: adding $U(z) - U(x)$ cancels the source. Dismissed.** Under $\tau = A^2/N^2 \rightarrow 0$, its gradient and Hessian are lower order while the joint source (3.6) persists.
4. **Objection: the no-go rules out every Bellman value function. Upheld as false.** It applies only to the bounded-coefficient, cutoff-summable finite-bank local class defined in Section 5. The nonlocal adapted-control value (8.3) is outside its scope.
5. **Objection: the Riesz inequality alone closes the full joint gate. Upheld as false.** It closes a frozen coherent-ray budget; the coefficient source adds the separate factor four, and general commutators remain under (8.4).
6. **Objection: (7.4) may be summed shell by shell. Upheld as invalid.** Its $H_A(x)$ contains Gaussian past energy. A11 proves that the corresponding repeated payment diverges with cutoff.
7. **Objection: the quartic destroys potential coercivity. Dismissed.** Equation (7.5) is the exact one-variable optimisation. On $L = 16$ its additive constant is $41.3631905705\dots$
8. **Objection: this package proves A7 Nelson or an interacting measure. Upheld as false.** Those conclusions require (8.4), which remains open. The tier stays T4.

Reproduction

Run from the repository root:

```
python codes/foundations/a13_classii_joint_source_potential_reduction_verify.py
```

The command runs the primary coefficient/Fierz route, a non-importing alias-free/Gauss–Hermite route, and integrated hash and boundary checks.

Result footer

Result ID: A13-CLASSII-JOINT-SOURCE-POTENTIAL-REDUCTION
Precise statement: Exact joint completion and asymptotic factor-four source refute coefficient-one conditioning and the precisely scoped finite-bank local Bellman class.
Scope: L=16 fixed-floor A1 production coefficients; six-real fields; strict sharp-cube filtration.
Dependencies: A1-PRODUCTION-FUNCTIONAL-REALISATION; A7--A13 Class-II chain (full IDs in card).
Evidence grade: ANALYTIC, EXACT, EXECUTED.
Reproduction command: python codes/foundations/a13_classii_joint_source_potential_reduction_verify.py
Expected output: A13-CLASSII-JOINT-SOURCE-POTENTIAL-REDUCTION-INTEGRATED-PASS; 105/105; exit 0.
Falsification gate: Any failed completion, doubling, mixed-Hardy, one-shell, cross-route, hash, or PDF check.
Tier before / after: T4 / T4
No-overclaim statement: No A7 Nelson bound, interacting measure, T5/T6/T7, floor removal, infinite volume, phase transition, or BCC selection
Next required action: A13-CLASSII-CONTROLLED-SHELL-ENERGY-ONE-USE.

References

1. M. Boue and P. Dupuis, “A variational representation for certain functionals of Brownian motion,” *Annals of Probability* 26 (1998), 1641–1659, <https://doi.org/10.1214/aop/1022855876>.
2. N. Barashkov and M. Gubinelli, “A variational method for Φ_3^4 ,” <https://arxiv.org/abs/1805.10814>.
3. B. Hollenbeck and I. E. Verbitsky, “Best constants for the Riesz projection,” *Journal of Functional Analysis* 175 (2000), 370–392, <https://doi.org/10.1006/jfan.2000.3616>.
4. TECT A7–A13 claim packages, exact covariance-normal composite, determinant, true-increment, and relative-phase obstruction records.